



BUSINESS TRACK

REDUCING RTO: GETTING MORE ORDERS DELIVERED

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Problem Statement Redefined

Objective

Reduce RTO while preserving Meesho's COD-led growth and improving Valmo's last-mile economics.

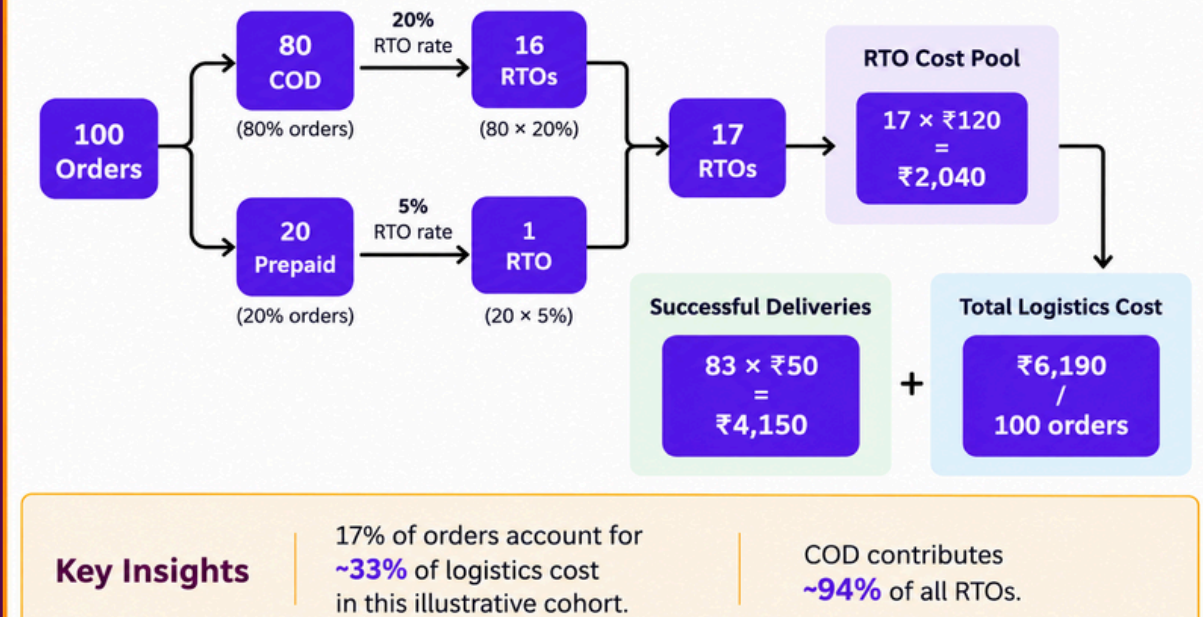
Potential Opportunity

- Convert RTO from a pure cost centre into a controllable operational metrics
- 94% of RTOs come from COD in the case baseline
- Risk based prepaid conversion can reduce expected RTO cost
- Rider proof of attempt can reduce operationally induced RTO
- Local RTO interception can avoid unnecessary reverse transportation
- Valmo's decentralized LMDC network creates an opportunity for local resale

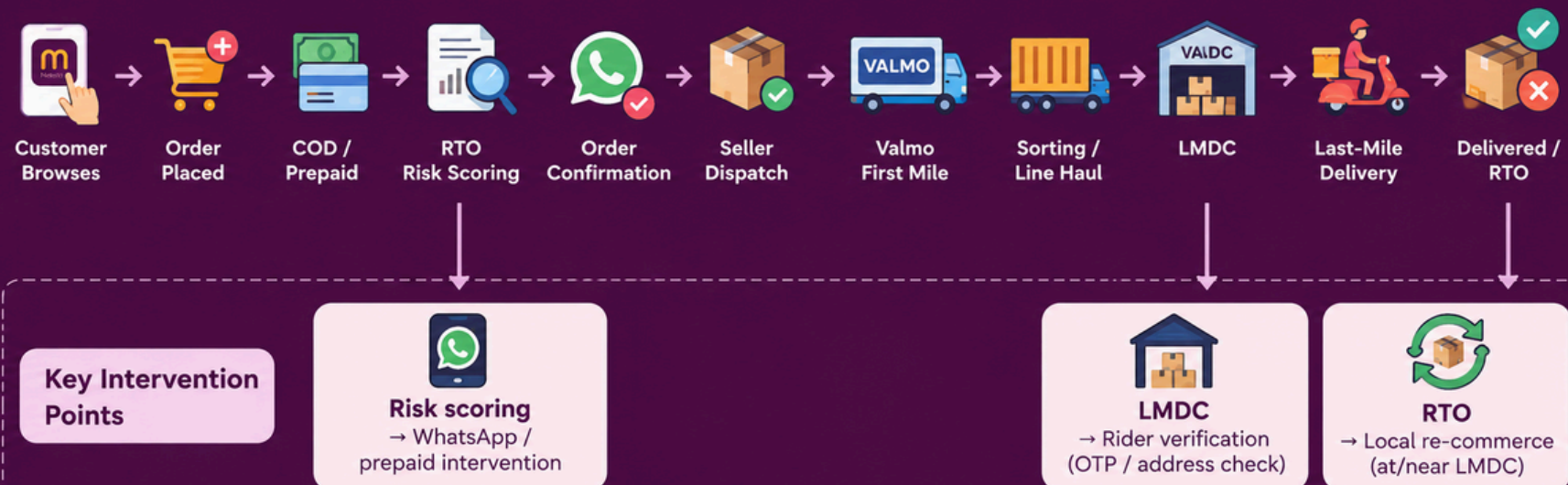
Challenges

- **Customer Intent:** COD dominates orders (80%); buyer remorse and temporary cash shortages contribute to delivery refusal.
- **Last Mile Operations:** Unstructured addresses and difficult locations increase RTO; deliveries >10 km from LMDC reach 22% RTO.
- **Rider Incentives:** Low payout (~₹21/delivery) can create incentive misalignment for high-effort deliveries and potential fake attempts.
- **Reverse Logistics:** RTO costs ₹120 vs ₹50 for successful delivery ; a 2.4× cost asymmetry that significantly impacts margins.

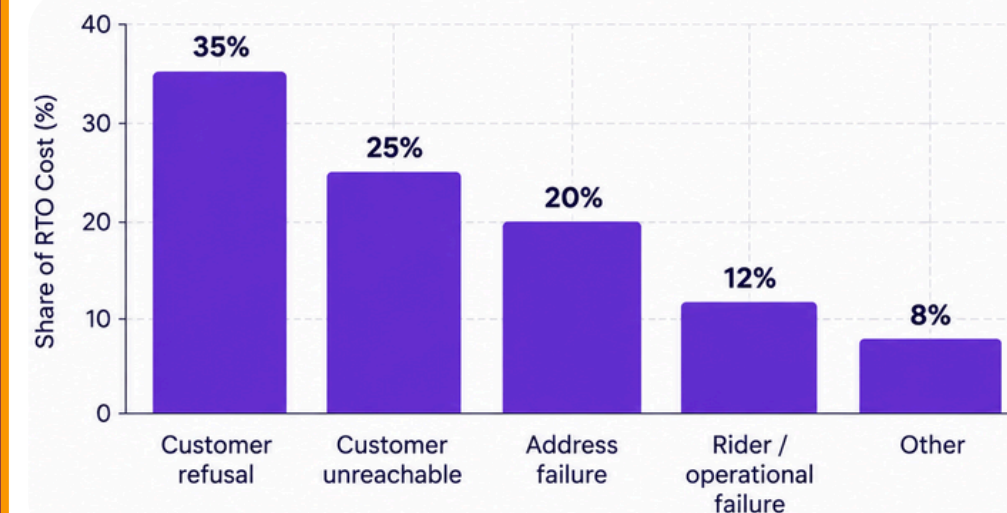
Market Idea:



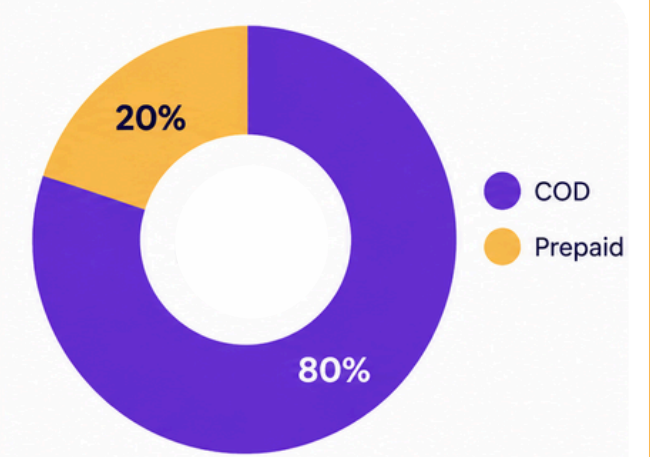
Meesho-Valmo Order Flow



RTO Failure Driver (%)



Payment Method (%)



Delivery Stakeholder Personas

1. Last-Mile Rider Profile: Valmo delivery partner serving an LMDC. Goals: Maximize earnings, complete deliveries efficiently, minimize travel/time. Pain Points: • Long distance routes • Unstructured addresses • Customer unavailable • Low reward relative to effort Digital Maturity: High Opportunity for Meesho: GPS proof of attempt + route optimization + differentiated incentives for difficult deliveries.	2. COD Customer Profile: First-time / infrequent Meesho customer choosing COD. Goals: Convenience, no upfront payment, easy returns. Pain Points: • Buyer remorse • Cash unavailable at delivery • Doesn't want to wait 3–5 days • Address/location ambiguity Digital Maturity: High Opportunity for Meesho: Targeted prepaid incentive + WhatsApp confirmation + live location sharing.	3. LMDC / Hub Operator Profile: Local logistics node handling shipment sorting and last mile allocation. Goals: High throughput, low RTO, efficient rider utilization. Pain Points: • Poor address quality • Rider capacity mismatch • High failed-attempt volumes • RTO inventory accumulation Digital Maturity: High Opportunity for Meesho: RTO heatmaps + rider analytics + local recommerce inventory matching.	4. Seller / Merchant Profile: Merchant whose product gets returned. Goals: Successful delivery, minimize inventory loss, protect margins. Pain Points: • ₹120 RTO hit • Restocking cost • Packaging damage • Inventory tied up during reverse journey Digital Maturity: Moderate to High Opportunity for Meesho: Local RTO interception + faster inventory recovery + merchant visibility.
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Competitive Benchmarking

Platform	COD/RTO Approach	Key Initiatives	Last-Mile Model	Customer Intervention	Reverse Logistics	Unmet Gap / Meesho Opportunity
Meesho / Valmo	COD; tech led multi partner network.	Route optimization; local partners.	Partners, hubs, sorting & 3PLs.	Pre dispatch checks; confirmation; interception.	Local interception & rerouting.	Link RTO prediction with post-refusal interception.
Amazon	COD/Pay on Delivery; screening & NDR.	FBA, Easy Ship, tracking, OTP.	FBA, Easy Ship & seller fulfilment.	Tracking, notifications, OTP, contact.	Managed returns.	Differentiate via decentralized local interception.
Flipkart	COD/prepaid; COD seller fee.	Authentication, communication, OTP.	Ekart & partners.	Updates, communication, OTP.	Returns & reverse logistics.	Predictive pre dispatch + refusal interception.
D2C logistics players	COD/prepaid; RTO reduction.	NDR, analytics, orchestration, RTO prediction.	Multi carrier / 3PL.	IVR, WhatsApp, SMS, NDR, reattempts.	RTO & reverse movement.	Few control the full prediction-to-interception journey.

Models to Reduce RTO & Recover Delivery Value

MODEL 1 — RTO RISK ENGINE + PRE-DISPATCH INTERVENTION

What it is

A real-time model predicts the RTO probability of every COD order and triggers the cheapest effective intervention based on the risk level.

Execution Flow:

- Order placed → COD order enters the RTO risk engine.
- RTO risk score generated using customer, order, address and historical delivery signals.
- Low risk → Normal COD journey.
- Medium risk → WhatsApp confirmation + address/location verification.
- High risk → ₹15 prepaid incentive / mandatory order confirmation.
- Confirmed order → Seller dispatch → Valmo delivery.
- Failed intervention → Order held/cancelled, avoiding an unnecessary ₹120 RTO cost.

Benefits:

- Targets high risk COD orders instead of adding friction to all customers.
- Converts selected COD orders to prepaid, reducing RTO exposure.
- Improves address quality and first attempt delivery success.
- Preserves COD accessibility for reliable customers.
- Makes interventions economically viable by comparing ₹15 intervention cost vs. ₹120 expected RTO loss.

Bottlenecks and Solutions:

- False positives → Continuously track intervention conversion, cancellations and customer complaints.
- ₹15 incentive economics → Trigger only when expected RTO loss > intervention cost.
- Customer friction → Use single click WhatsApp confirmation with minimal additional steps.
- Model bias / cold start → Retrain using actual RTO outcomes and use broader order/address signals where customer history is limited.
- Intervention failure → Escalate only high-risk orders to stronger confirmation or prepaid conversion.



Critical Thinking & Fit

Preventive Layer: Model 1 reduces RTO through targeted pre dispatch intervention.

Recovery Layer: Model 2 reduces the economic impact of unavoidable RTO through local recommerce.

Strategic Fit: Both leverage COD data, Valmo's LMDC network, WhatsApp, and local delivery infrastructure.

Model 2 – RTO Interception + Local Re-commerce

What it is:

Instead of automatically sending refused shipments back through the reverse network, eligible products remain at the destination LMDC and are matched with nearby demand.

Execution Flow:

- Customer refuses delivery.
- Shipment reaches the destination LMDC.
- Product eligibility and condition check.
- Nearby customer demand identified.
- Match found → Offer targeted 15–20% discount.
- Product delivered to a new nearby customer.
- No match → Proceed with normal RTO.

Benefits:

- Eliminates unnecessary reverse line haul.
- Reduces inventory transit and handling.
- Recovers value from refused shipments.
- Converts RTO inventory into local inventory.
- Potentially reduces the ₹120 RTO cost.

Bottlenecks and Solutions:

- Product damaged/opened → Eligibility inspection before relisting.
- No local demand → Route the shipment through the normal RTO process.
- Merchant doesn't want resale → Make local recommerce merchant opt in.
- Customer distrust → Maintain Meesho's existing quality and review layer.
- Inventory mismatch → Use local demand prediction + SKU matching.



KPIs for Success

Model 1 → **RTO ↓** • COD→Prepaid • Confirm % • 1st-attempt delivery % • RTO cost avoided • Net savings/order
 Model 2 → **RTO intercepted %** • Re-commerce % • Recovery/RTO • Reverse cost avoided • Delivery success % • Net savings/RTO